When f^2 Stays Near 1: The Real Part of f Has Constant Sign

Colin McConnell

Source: L. V. Ahlfors, Complex Analysis, 3rd ed., p. 72, Exercise 3 (MIT OCW 18.112, Problem Set 2)

Problem Statement. Suppose that $f(z)$ is analytic and satisfies $|f(z)^2 - 1| < 1$ in a region Ω. Show that either $\operatorname{Re} f(z) > 0$ throughout Ω or $\operatorname{Re} f(z) < 0$ throughout Ω.

Recall that a *region* is a nonempty open connected subset of $\mathbb{C}$.

Solution

The proof has two steps. First, the hypothesis rules out $\operatorname{Re} f(z) = 0$ at every point. Second, connectedness prevents $\operatorname{Re} f$ from taking both signs.

Proof. Step 1: $\operatorname{Re} f$ *never vanishes.* Suppose $\operatorname{Re} f(z_0) = 0$ for some $z_0 \in \Omega$. Then $f(z_0) = iy$ for some $y \in \mathbb{R}$, so $f(z_0)^2 = -y^2$ and

$$|f(z_0)^2 - 1| = |-y^2 - 1| = y^2 + 1 \geq 1.$$

This contradicts the hypothesis $|f(z_0)^2 - 1| < 1$. Hence $\operatorname{Re} f(z) \neq 0$ for every $z \in \Omega$.

Step 2: the sign is constant. Since f is analytic, it is continuous, and therefore $u = \operatorname{Re} f$ is a continuous real-valued function on Ω. Define

$$A = \{z \in \Omega : u(z) > 0\}, \qquad B = \{z \in \Omega : u(z) < 0\}.$$

Then:

- A and B are open, since they are the preimages under the continuous function u of the open sets $(0, \infty)$ and $(-\infty, 0)$, and Ω is open;
- $A \cap B = \varnothing$;
- $A \cup B = \Omega$, by Step 1.

Since Ω is connected, it cannot be written as the union of two disjoint nonempty open sets. Therefore $A = \varnothing$ or $B = \varnothing$. If $B = \varnothing$, then $\operatorname{Re} f(z) > 0$ throughout Ω. If $A = \varnothing$, then $\operatorname{Re} f(z) < 0$ throughout Ω. $\square$